Deep-hole rigidity of the Clebsch hexagon code

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July 2026

Abstract

Among the six-arcs of $\mathrm{PG}(2, 11)$, exactly one projective class has the maximum-distance syndrome directions of its associated MDS code contained in a conic: the Clebsch hexagon. This coding-theoretic condition recovers its projective stabilizer A_5. More generally, containment of the syndrome locus in the zero set of any nonzero form of degree at most three characterizes the same class. A line bound, a chord-defect identity, and Dye's Brianchon theorem give the core conic-rigidity implication. Enumeration proves the low-degree strengthening, the sharp numerical gap, and the terminal $k = 7$ exclusions.

The arc lies on no conic, so its $[6, 3, 4]_{11}$ MDS code is non-GRS; the code has covering radius three. Its twelve projective deep-hole directions are the $\mathbb{F}_{11}$-points of a conic. Among nonzero syndromes outside the six arc columns, the quadratic test separates exactly the deep holes; together with the zero-syndrome and arc-column cases, it gives the complete distance oracle of Proposition 3.5. Its ambiguity data recover the classical Brianchon points, self-polar triangles, and an invariant support bipartition. The chord identity and the Sylvester graph show that $q = 11$ is the only conic-filling field for six-arcs. A universal chord-defect identity and passant count show that conic filling by a k-arc forces q odd and $2k - 3 \le q < \binom{k}{2}$, with a sharper quadratic barrier. Through seven points the only cases are a four-frame in $\mathrm{PG}(2, 5)$ and the Clebsch hexagon; for eight points only $q \in \{13, 17, 19\}$ remain.

Keywords. Clebsch hexagon; MDS code; deep hole; finite projective plane; arc; conic.

2020 Mathematics Subject Classification. 51E21, 94B05, 51E22, 05B25.

1 Introduction

In $\mathrm{PG}(2, 11)$ there is a six-point set whose fifteen chords miss exactly twelve points, and those twelve points are the points of a conic. This paper shows that the property characterizes the set: it is the Clebsch hexagon. In coding language, the conic is the projective deep-hole syndrome locus of a non-GRS $[6, 3, 4]_{11}$ MDS code. This is an instance of a broader question: when does the projective deep-hole locus determine an MDS code, and when can that locus be algebraically degenerate?

Let C be a linear $[n, k, d]_q$ code with covering radius ρ and full-rank parity-check matrix H. For a received word v, write $\sigma(v) = Hv^{\mathsf{T}}$ and

$$w(s) = \min\{\mathrm{wt}(e) : He^{\mathsf{T}} = s\}.$$

Then $d(v, C) = w(\sigma(v))$, and v is a *deep hole* precisely when this minimum is ρ. The syndrome s specifies one coset of C, whose coset leaders are its weight-$w(s)$ representatives. Because $w(\lambda s) = w(s)$ for $\lambda \ne 0$, maximum-distance syndrome directions form a projective locus

$$\mathcal{D}_{\mathrm{proj}}(C) = \{[s] \in \mathrm{PG}(n - k - 1, q) : w(s) = \rho\}.$$

Thus one projective direction represents $q-1$ distinct nonzero syndrome cosets, while each coset contains $|C|$ received words.

For an MDS code of redundancy $r = n-k$, the parity-check columns form a projective arc in $\mathrm{PG}(r-1, q)$, and $w(s)$ is the least number of columns whose span contains $[s]$. Adjoining a new column preserves the MDS property exactly when its direction avoids every hyperplane spanned by $r-1$ existing columns. In the redundancy-three plane case used here, this specializes to weights one, two, or three according as the direction lies on the arc, on a secant of the arc, or on neither; an extension point is precisely a direction missed by every secant. We write $\mathcal{U}(A)$ for this extension locus. If $\mathcal{U}(A) \neq \emptyset$, then the associated code has covering radius three and $\mathcal{U}(A) = \mathcal{D}_{\mathrm{proj}}(C)$. This dictionary is classical in the theory of complete arcs and is developed for MDS codes by Davydov, Marcugini, and Pambianco [7]; their Theorem 6.3 supplies the leader count used in Section 3. See also Hirschfeld [14] for the plane arc/covering-radius setting.

The main theorem starts from no symmetry or conic construction. Among all six-arcs of $\mathrm{PG}(2, 11)$, conic containment of the projective deep-hole syndrome locus reconstructs the Clebsch hexagon and its A_5 stabilizer. A line bound, a chord-defect identity, and Dye's Brianchon theorem give the conceptual implication; the census is needed only for the numerical clause. We then prove quantitative gaps, reconstruct the Brianchon geometry from decoder ambiguity, and isolate $q = 11$. For a Clebsch hexagon K, the same chord identity gives $|\mathcal{U}(K)| = q^2 - 14q + 45$, with off-conic excess $(q-4)(q-11)$ when $q \equiv 3 \pmod 4$. In general the same moment calculation shows that a conic-filling k-arc must satisfy q odd and $2k - 3 \leq q < \binom{k}{2}$, together with a sharper quadratic barrier; we classify the cases through seven points and reduce the eight-point case to three fields.

The configuration goes back to Clebsch [3, p. 336]. Edge constructed its finite-plane form and determined its order-60 stabilizer [4, §§29–32]; Dye proved projective uniqueness and determined the stabilizer over a general field [12, Theorems 1 and 3, pp. 275–278]; Storme and Van Maldeghem identified the corresponding six-arc in their finite-plane classification [18]. Deep holes of Reed–Solomon codes have both an algorithmic history [13, 6] and a geometric classification in the projective case [19]. Those results concern GRS codes, whose arcs lie on normal rational curves; the code here is not GRS.

What is new and what is not. The hexagon, its uniqueness and stabilizer, and its complete-exterior-set interpretation are classical [3, 4, 12, 1, 18]. Sadeh classified the six-arcs of $\mathrm{PG}(2, 11)$ [16]; we claim no priority for that census or its extension counts. We prove the conic-locus identification of Proposition 3.2 directly here. The new contribution is the symmetry-free coding criterion and its consequences: low-degree rigidity, quantitative gaps, leader-support bipartition, decoder reconstruction, and the finite-field boundary.

2 The Clebsch hexagon

Fix the nonsingular conic

$$\mathcal{C}: \ XZ = Y^2$$

in $\mathrm{PG}(2, 11)$, with its 12 points identified with $\mathrm{PG}(1, 11) = \mathbb{F}_{11} \cup \{\infty\}$ in the usual way. The stabilizer of $\mathcal{C}$ in $\mathrm{PGL}(3, 11)$ is $\mathrm{PGL}_2(11)$, acting on $\mathcal{C}$ as it acts on the projective line.

A convenient representative is

$$A = \{(1, 10, 0),\ (1, 9, 1),\ (1, 4, 7),\ (1, 8, 5),\ (0, 1, 4),\ (1, 1, 7)\}.$$

It is a six-arc disjoint from $\mathcal{C}$; its fifteen joins also avoid $\mathcal{C}$. The following invariant construction explains its symmetry.

Definition 2.1 (Clebsch hexagon)**.** Let $A_5 \subset \mathrm{PGL}_2(11)$ be an icosahedral subgroup. The six Sylow 5-subgroups of A_5 each fix two points of $\mathcal{C}$. The *Clebsch hexagon* is the set of poles of the six chords joining these fixed pairs.

For a nonsingular conic over an odd field, an *external point* lies on two tangents and an *internal point* on none; an external, or *passant*, line is disjoint from the conic. Edge's six vertices are external, their fifteen joins are external, and those joins concur three at a time at ten internal Brianchon points [4, §§29–32]. Thus the hexagon is a complete exterior set in the terminology of Blokhuis, Seress, and Wilbrink [1, pp. 143,146]. Their $q = 11$ list also contains a Pasch configuration, but its four collinear triples prevent it from defining an MDS code. Thus exteriority alone gives only $\mathcal{C}(\mathbb{F}_{11}) \subseteq \mathcal{U}(A)$; the arc hypothesis selects the Clebsch branch, and Proposition 3.2 proves equality.

The displayed hexagon is the arc K_2 of Storme and Van Maldeghem [18, Props. 11–12] and Dye's synthetic hexagon when 5 is a square in the base field [12, Theorem 1]. Dye proves uniqueness, stabilizer A_5, and the unique associated polarity; in the displayed coordinates its conic is $\mathcal{C}$. His calculation also shows that the six vertices lie on no conic in characteristic 11 [12, Theorem 2, pp. 277–278; p. 281]. Storme and Van Maldeghem record that the arc is incomplete at $q = 11$ [18, Prop. 13], so the associated redundancy-three code has covering radius three.

Theorem 2.2 (Universal chord defect)**.** *Let A be a k-arc in $\mathrm{PG}(2, q)$ with $k \geq 4$. Let n_i count the off-arc points incident with exactly i chords, put*

$$r = \lfloor k/2 \rfloor, \qquad t(A) = \sum_{i=3}^{r} \binom{i-1}{2} n_i,$$

and write $m = \binom{k}{2}$. Then

$$|\mathcal{U}(A)| = q^2 - (m-1)q + \left(m + 3\binom{k}{4} - k + 1\right) - t(A), \qquad 0 \leq t(A) \leq 3\binom{k}{4}\frac{r-2}{r}.$$

For $k \leq 5$, the defining sum is empty and $t(A) = 0$.

Proof. At a point off A, concurrent chords have pairwise disjoint endpoint pairs, so at most r chords occur. Counting chord–point incidences and pairs of disjoint chords gives

$$\sum_i i n_i = m(q-1), \qquad \sum_i \binom{i}{2} n_i = 3\binom{k}{4}.$$

Since $\binom{i}{2} = (i-1) + \binom{i-1}{2}$, the number of covered off-arc points is $m(q-1) - 3\binom{k}{4} + t(A)$. Subtraction from the $q^2 + q + 1 - k$ off-arc points proves the identity. Finally,

$$\binom{i-1}{2} \leq \frac{r-2}{r}\binom{i}{2} \qquad (3 \leq i \leq r),$$

and the second moment gives the defect bound. □

Corollary 2.3 (Conic-filling window and six-arcs)**.** *If $|\mathcal{U}(A)| = q + 1$, then*

$$t(A) = q^2 - mq + m + 3\binom{k}{4} - k, \qquad q^2 - mq + m - k + \frac{6}{r}\binom{k}{4} \leq 0,$$

and $q < m$. If q is even, then $\mathcal{U}(A)$ is not an arc. Hence if $\mathcal{U}(A)$ is a nonsingular conic, then q is odd and

$$2k-3 \le q < \binom{k}{2}.$$

For $k = 6$, writing $c(A) = n_3$ gives

$$|\mathcal{U}(A)| = q^2 - 14q + 55 - c(A), \qquad 0 \le c(A) \le 15, \qquad c(A) = (q-6)(q-9)$$

when $|\mathcal{U}(A)| = q + 1$. For the displayed Clebsch hexagon, Dye's ten Brianchon points give $c(A) = 10$; its fifteen chords have 45 disjoint pairs, cover 115 of the 127 off-arc points, and leave exactly twelve uncovered.

Proof. The first formula is Theorem 2.2 rearranged, and its comparison with the defect bound gives the quadratic inequality. There are $q^2 - k$ covered off-arc points, so $q^2 - k \le m(q-1)$; if $q \ge m$, then $0 \le q(q-m) \le k - m < 0$, a contradiction.

Suppose q is even and $\mathcal{U}(A)$ is an arc. Since it has $q+1$ points, it has a nucleus $N \notin \mathcal{U}(A)$ through which every line is tangent [14, p. 177]. Every chord of A is disjoint from $\mathcal{U}(A)$, so no chord contains N; nor can N be a vertex. Thus $N \in \mathcal{U}(A)$, a contradiction.

In the conic case, q is therefore odd and every chord is passant. An off-conic point lies on at most $(q+1)/2$ passants, so the $k-1$ chords through a vertex give $q \ge 2k-3$.

For $k = 6$, one has $r = 3$ and $t(A) = n_3 = c(A)$, giving the displayed specialization. For the displayed arc, Dye's Brianchon theorem gives $c(A) = 10$ [12, Theorem 1, pp. 275–276]. At $q = 11$, the two moments read $\sum i n_i = 150$ and $\sum \binom{i}{2} n_i = 45$, whence $n_1 + n_2 + n_3 = 115$. □

3 The code and its projective syndrome locus

Take the parity-check matrix

$$H = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 1 \\ 10 & 9 & 4 & 8 & 1 & 1 \\ 0 & 1 & 7 & 5 & 4 & 7 \end{pmatrix},$$

whose columns are the six displayed points of A, and let

$$C = \{c \in \mathbb{F}_{11}^6 : Hc^{\mathsf{T}} = 0\}$$

be the resulting $[6,3,4]_{11}$ MDS code. Since the six columns of H do not lie on any conic (their evaluation matrix against the six quadratic monomials $X^2, XY, XZ, Y^2, YZ, Z^2$ is nonsingular), C is not monomially equivalent to a GRS code: the dual of a GRS code is GRS, and the parity-check column system of a length-six, dimension-three GRS code lies on a normal rational curve, here a conic, in the classical normal-rational-curve/GRS dictionary [17].

The arc–coset dictionary from the introduction applies without a GRS hypothesis [7, Def. 6.1, Thm. 6.3]. Each uncovered direction x lifts to $q-1$ weight-three cosets, and each has $\binom{6}{3} = 20$ minimum-weight leaders. Geometrically, the same condition says that adjoining x as a seventh column preserves the MDS property.

For example, $s = (1, 0, 0)$ lies on $\mathcal{C}$ and is therefore uncovered. The error

$$e = (7, 4, 1, 0, 0, 0)$$

satisfies $He^{\mathsf{T}} = s$. No error of weight one or two has syndrome s, while the twenty triples of coordinate positions each support one weight-three leader. Thus this single syndrome exhibits the complete dictionary: its projective direction is a conic point, its coset has distance three, and its nearest-codeword ambiguity has size twenty.

The stabilizer already forces the small point sets that will occur.

Proposition 3.1 (The A_5 point orbits)**.** *Let $G = \mathrm{Stab}(A) \cong A_5$. Its orbits on* $\mathrm{PG}(2, 11)$ *have lengths*

$$6,\ 10,\ 12,\ 15,\ 30,\ 30,\ 30.$$

The first four are respectively A, the ten Brianchon points, $\mathcal{C}(\mathbb{F}_{11})$, and the fifteen vertices of the five self-polar triangles. In particular, $\mathcal{C}(\mathbb{F}_{11})$ is the unique twelve-point G-orbit.

Proof. Dye's polarity and stabilizer theorems identify G with the ternary icosahedral representation preserving $\mathcal{C}$, and his Corollary 1 makes it transitive on the six vertices, ten Brianchon points, and fifteen triangle vertices [12, Theorems 2–3 and Corollary 1, pp. 276–279]. We derive the orbit sizes without using the finite enumeration.

The argument first determines the points with each possible nontrivial stabilizer, then applies orbit–stabilizer to recover the orbit lengths.

Since $11 \nmid 60$, the representation is semisimple, and the ternary icosahedral representation is irreducible; hence G fixes no projective point. The ordinary three-dimensional icosahedral character, reduced in $\mathbb{F}_{11}$, shows that an involution, an element of order three, and an element of order five fix respectively 13, 1, and 3 projective points. Indeed, an involution has a one-dimensional eigenspace and a two-dimensional eigenspace; the nontrivial order-three eigenvalues are not in $\mathbb{F}_{11}$; and the three order-five eigenvalues are distinct because $5 \mid 10$. Restricting the same representation to the standard subgroups of A_5 gives the following common-fixed-point ledger; the standard subgroup classification of A_5 shows that these are all possible nontrivial proper point stabilizers. Here D_5 has order ten, and the last row counts points whose stabilizer is exactly the displayed subgroup type. For the noncyclic entries, determine the common projective eigenlines of standard generators: a D_5 normalizer retains one of the three C_5 eigenlines, an S_3 normalizer retains the unique C_3 eigenline, a V_4 has three simultaneous eigenlines, and its overgroup A_4 permutes those three and fixes none. This gives the second row.

H	A_4	D_5	S_3	C_5	V_4	C_3
number of conjugate subgroups	5	6	10	6	5	10
fixed points for one H	0	1	1	3	3	1
points with stabilizer conjugate to H	0	6	10	12	15	0

For the third row, a D_5 fixes one C_5 eigenline and exchanges the other two; the unique C_3 line is already fixed by its S_3 normalizer; and A_4 permutes the three V_4 eigenlines without fixing one. By orbit–stabilizer, the third row already gives the orbits of lengths 6, 10, 12, and 15; only points with exact stabilizer C_2 remain. It remains to account for exact C_2 stabilizers. Each involution lies in two D_5 subgroups, two S_3 subgroups, and one V_4. Of its thirteen fixed points, these contribute respectively 2, 2, and 3 points from the preceding rows. The remaining six points have exact stabilizer C_2. There are fifteen involutions, so this gives 90 points, necessarily three orbits of length $60/2 = 30$. The total $6 + 10 + 12 + 15 + 90 = 133$ exhausts the plane. Dye's Theorem 6(v), specialized at $q = 11 \equiv 1 \pmod{10}$, identifies the twelve-point C_5-stabilizer orbit with $\mathcal{C}(\mathbb{F}_{11})$ [12, pp. 281–282; §3.1, p. 284]; the other three identifications are the transitive sets cited above. □

Proposition 3.2 (The uncovered locus is the conic)**.** *The covering radius of C is three, and*

$$\mathcal{U}(A) = \mathcal{D}_{\mathrm{proj}}(C) = \mathcal{C}(\mathbb{F}_{11}).$$

Proof. Corollary 2.3 gives $|\mathcal{U}(A)| = 12$. This set is G-invariant and disjoint from the six-point orbit A, so Proposition 3.1 identifies it with the unique twelve-point orbit $\mathcal{C}(\mathbb{F}_{11})$. The arc–coset dictionary identifies this set with the projective deep-hole syndrome locus, and its nonemptiness gives covering radius three. □

Corollary 3.3 (Directions, cosets, leaders, and received words)**.** *$\mathcal{U}(A)$ has 12 directions, exactly the points of $\mathcal{C}$. Above them lie $12(11-1) = 120$ nonzero maximum-distance syndromes, hence 120 deep-hole cosets. Each such coset has $\binom{6}{3} = 20$ minimum-weight leaders, for 2400 leaders in all, and contains $|C| = 11^3 = 1331$ received words. Consequently C has $120 \cdot 1331 = 159720$ received-word deep holes.*

Proof. Immediate from Proposition 3.2, the leader-count clause of the DMP dictionary, and the fact that the fiber of $\mathbb{F}_{11}^3 \setminus \{0\} \to \mathrm{PG}(2, 11)$ over each direction consists of ten nonzero syndromes, hence ten distinct cosets. Every coset has $|C|$ words. □

3.1 Automorphisms and deep-hole orbits

The projective stabilizer of the six column rays is A_5 of order 60, acting faithfully and 2-transitively on the coordinate supports. It is the support-permutation quotient of the monomial automorphism group:

$$1 \longrightarrow \mathbb{F}_{11}^{\times} \longrightarrow \mathrm{MAut}(C) \longrightarrow A_5 \longrightarrow 1.$$

The kernel consists of the ten global coordinate scalars. The sequence splits. Since $\gcd(3, 10) = 1$, each projective class has a unique determinant-one representative; their monomial lifts give an A_5 complement. Hence

$$\mathrm{MAut}(C) \cong C_{10} \times A_5$$

has order 600. For the displayed column representatives, the subgroup of pure coordinate permutations is trivial.

Proposition 3.4 (One orbit on received-word deep holes)**.** *The monomial automorphism group of C is transitive on the 120 deep-hole cosets. Let Γ be the group generated by these automorphisms and by translations $v \mapsto v + c$ for $c \in C$. Then*

$$|\Gamma| = |C|\,|\mathrm{MAut}(C)| = 1331 \cdot 600 = 798600,$$

and Γ is transitive on all 159720 received-word deep holes. The stabilizer of each such word is cyclic of order five.

Proof. The projective stabilizer A_5 is transitive on the twelve points of $\mathcal{C}$. Indeed, its six Sylow 5-subgroups form one conjugacy class, each fixes the two endpoints of one of the six fixed-point chords, and the involution in its dihedral normalizer interchanges those endpoints. These twelve endpoints exhaust $\mathcal{C}(\mathbb{F}_{11})$. Every such projectivity lifts to a monomial automorphism of the parity-check columns, so the monomial group is transitive on the twelve syndrome rays above $\mathcal{C}$. Its central scalar subgroup $\mathbb{F}_{11}^{\times}$ acts transitively on the ten nonzero vectors of each ray. Hence the monomial group is transitive on all $12 \cdot 10 = 120$ maximum-distance syndromes and therefore on the corresponding cosets. Given deep holes v and w, choose $m \in \mathrm{MAut}(C)$ with $\sigma(mv) = \sigma(w)$. Then $c = w - mv$ has zero syndrome, so $c \in C$, and the codeword translation by c sends mv to w. Both kinds of generator are Hamming isometries preserving C. The monomial group normalizes the translation subgroup, since $m\,t_c\,m^{-1} = t_{m(c)}$ with $m(c) \in C$. Thus $\Gamma = C \rtimes \mathrm{MAut}(C)$; the two subgroups have trivial intersection and the displayed product order follows. Orbit–stabilizer now gives stabilizer order $798600/159720 = 5$, and every group of prime order is cyclic. □

3.2 Decoding consequences

The coset-leader-weight distribution of C is $(1, 60, 1150, 120)$ (cosets of weight $0, 1, 2, 3$, summing to $11^3 = 1331$); the last entry counts deep-hole cosets, not received words. Its codeword weight enumerator is $(1, 0, 0, 0, 150, 420, 760)$, forced by the MDS parameters. This is the radius-three instance of the coset-weight framework of Davydov, Marcugini, and Pambianco [7, §6]; the secant indices below refine its weight-two stratum by nearest-word multiplicity.

The special geometry also makes distance diagnosis and nearest-word structure unusually explicit. Write a syndrome as $s = (s_0, s_1, s_2)$ and put $Q(s) = s_0 s_2 - s_1^2$. For a point $x \notin A$, its *secant index* is the number of chords of A through x; by the dictionary above, this is the number of weight-two leaders for each nonzero syndrome on the ray x.

Proposition 3.5 (Complete syndrome oracle and ambiguity enumerator)**.** *For $v \in \mathbb{F}_{11}^6$ with syndrome $s = Hv^{\mathsf{T}}$,*

$$d(v, C) = \begin{cases} 0, & s = 0, \\ 1, & [s] \in A, \\ 3, & s \neq 0 \textit{ and } Q(s) = 0, \\ 2, & \textit{otherwise}. \end{cases}$$

Among the 1330 *nonzero cosets, the numbers having respectively one, two, three, and twenty nearest codewords are*

$$960, \qquad 150, \qquad 100, \qquad 120.$$

For a distance-two syndrome the number of nearest codewords is the secant index of its direction. Every distance-three syndrome has exactly one leader on each of the twenty three-element coordinate supports.

Proof. The DMP dictionary identifies weight-one directions with the six columns, weight at most two with the secant union, and weight three with its complement. Proposition 3.2 identifies the last set with $Q = 0$, proving the oracle. In the notation of its proof, the chord-intersection equations give $(n_0, n_1, n_2, n_3) = (12, 90, 15, 10)$, where n_i counts projective directions of secant index i and n_0 is the uncovered locus. Every direction has ten nonzero affine syndromes. Thus the distance-two cosets contribute 900 cases of multiplicity one, 150 of multiplicity two, and 100 of multiplicity three; the sixty weight-one cosets add sixty more cases of multiplicity one. Leaders are in bijection with nearest codewords by translation within the coset. Finally, the three columns on any support are independent. For an uncovered syndrome their unique coefficients are all nonzero, since a zero coefficient would put the syndrome on a secant. Hence all twenty supports, and no others, give its weight-three leaders. □

3.3 Self-polar structure

Edge's finite-plane construction [4, §§30.2–31] and Dye's general-field synthetic theory [12, Theorem 2, pp. 277–278] exhibit five triangles on the six vertices, self-polar for $\mathcal{C}$. Their vertex-pair matchings partition the fifteen pairs into five perfect matchings of K_6. Section 5 turns its ten pairs of matchings into the Brianchon–support dictionary.

4 The rigidity theorem

Proposition 3.2 concerns one known arc. The following theorem is the paper's main claim: among *all* six-arcs of $\mathrm{PG}(2, 11)$, the property "the uncovered locus lies on a conic" picks out the Clebsch hexagon uniquely, and does so in a way that recovers its projective stabilizer from the coding condition rather than assuming it.

Two conditions, one word apart. The condition studied here is that $\mathcal{U}(A)$ — the projective uncovered, or projective deep-hole syndrome, locus — lies on a conic. It is *not* the familiar condition that the arc A itself lies on a conic, which is the standing hypothesis throughout the arc literature and which the Clebsch hexagon famously fails. The two are mutually exclusive for six-arcs in $\mathrm{PG}(2,11)$: Lemma 4.1 gives the exact four-value subcensus for six-arcs lying on a conic, so none belongs to the Clebsch class in clause (iii) of Theorem 4.2.

We first isolate the geometric input that removes degenerate conics from the computer-assisted part of the argument.

The external-line case is a six-point instance of the nonexistence in odd order of nontrivial *hyperfocused arcs*, whose secants determine only $k-1$ points on an external line [2, 5, 15]. We give a direct proof because the three possible line positions are then handled together.

Lemma 4.1 (Line bound and conic-inscribed subcensus)**.** *Let A be a six-arc in $\mathrm{PG}(2,q)$, where q is odd. Every line contains at most $q-5$ points of $\mathcal{U}(A)$. Moreover, if $q=11$ and A lies on a nonsingular conic, then*

$$|\mathcal{U}(A)| \in \{18, 19, 20, 22\}.$$

Proof. A chord contains no point of $\mathcal{U}(A)$. If a non-chord line ℓ contains one vertex, the ten chords on the other five vertices meet ℓ in at least five distinct points. Indeed, two of those chords sharing an endpoint meet only at that endpoint, which lies off ℓ; their intersections with ℓ therefore give a proper edge-colouring of K_5, whose chromatic index is five. Together with the vertex, at least six of the $q+1$ points of ℓ are therefore outside $\mathcal{U}(A)$.

It remains to consider $\ell \cap A = \varnothing$. The fifteen chords meet ℓ in covered points, and at most three chords pass through any one of them: chords sharing an endpoint meet only at that arc vertex, not on ℓ, so all chords through one point of ℓ have disjoint endpoint pairs. Hence there are at least five covered points. Equality would give a proper five-edge-colouring of K_6, necessarily a one-factorization. After relabelling, three factors are

$$01|23|45, \qquad 02|14|35, \qquad 03|15|24.$$

Send ℓ to infinity and normalize the first two directions to horizontal and vertical. Put $p_0=(0,0)$, $p_1=(x,0)$ and $p_2=(0,y)$, where $xy\neq 0$. Writing $p_3=(a,y)$, the first two factors force $p_4=(x,b)$ and $p_5=(a,b)$. Parallelism of 03 with 15 and with 24 gives two determinant equations whose difference is $2xy=0$, impossible for odd q. Thus a disjoint line has at least six covered points, and in all cases at most $(q+1)-6=q-5$ points of $\mathcal{U}(A)$ remain.

The final assertion is the conic-inscribed subcensus of the independent replay in the "Rigidity theorem" row of Table 3. □

Theorem 4.2 (Rigidity theorem)**.** *For a six-arc $A\subset \mathrm{PG}(2,11)$, the following are equivalent:*

(i) the uncovered locus $\mathcal{U}(A)$ lies on some conic (possibly degenerate);

(ii) $\mathcal{U}(A)$ is exactly the set of $\mathbb{F}_{11}$-points of a nonsingular conic;

(iii) A is $\mathrm{PGL}(3,11)$-equivalent to the Clebsch hexagon;

Proof. Suppose first that $\mathcal{U}(A)$ lies on a conic Q. If Q is nonsingular then $|\mathcal{U}(A)| \leq q+1 = 12$. If Q is degenerate, its rational points lie on one or two rational lines when it splits; in the nonsplit rank-two case its only rational point is the singular point. Thus Lemma 4.1 again gives $|\mathcal{U}(A)| \leq 12$. On the other hand, the chord-defect identity of Theorem 2.2 says

$$|\mathcal{U}(A)| = 22 - c(A),$$

where $c(A)$ is the number of nonvertex triple-edge concurrences. In Dye's terminology these are precisely the Brianchon points of the hexagon A. Over a field of characteristic different from two, Dye proves that every six-arc has at most ten Brianchon points and that the equality cases form a single $\mathrm{PGL}_3(K)$-orbit [12, §2.2–2.3, Theorem 1, pp. 275–276]. This bound has no symmetry, conic, or genericity hypothesis beyond the six-arc assumption. Hence $c(A) \leq 10$, so $|\mathcal{U}(A)| \geq 12$; equality holds throughout, $c(A) = 10$, and A is a Clebsch hexagon.

The degenerate-conic branch is therefore vacuous. Indeed, Proposition 3.2 and projective equivariance identify $\mathcal{U}(A)$ with the twelve rational points of a nonsingular conic. A degenerate conic cannot contain those same twelve points: it has no component in common with the nonsingular conic, so Bézout's theorem bounds their intersection by four. Thus the original containing conic is nonsingular. This proves (i)⇒(iii). Proposition 3.2 and projective equivariance give (iii)⇒(ii)⇒(i). Dye's stabilizer theorem also identifies the stabilizer in these equivalent cases with A_5. □

Table 1 prints the complete census at the level used by the rigidity and low-degree arguments. Every representative contains the standard frame

$$(0,0,1),\ (0,1,0),\ (1,0,0),\ (1,1,1);$$

the table gives the remaining two points of each representative. Here $d_{\min}$ is the least degree of a nonzero form vanishing on the uncovered locus.

In the second column, xy/uv abbreviates the remaining points $(1,x,y),(1,u,v)$.

class	pair	$\|G\|$	$\|\mathcal{U}\|$	$d_{\min}$
C01	23/32	2	20	5
C02	23/34	6	18	4
C03	23/36	3	19	5
C04	23/37	3	19	5
C05	23/39	4	20	5
C06	23/42	2	20	5
C07	23/48	1	21	5
C08	23/49	4	20	5
C09	23/56	4	20	5
C10	23/62	6	19	5
C11	23/67	12	16	4
C12	23/74	5	22	6
C13	23/89	6	18	4
C14	23/(9,10)	12	18	4
C15	34/45	60	12	2

Table 1: The fifteen projective classes of six-arcs in $\mathrm{PG}(2,11)$. Class C15 is the Clebsch class. Both replay programs regenerate the same canonical keys; the global-gap replay supplies $|G|$ and $|\mathcal{U}|$, and the low-degree replay supplies $d_{\min}$.

The same census supports a strengthening in which a cubic, rather than a conic, is allowed.

Proposition 4.3 (Low-degree algebraic rigidity)**.** *For a six-arc $A \subset \mathrm{PG}(2,11)$, the uncovered locus $\mathcal{U}(A)$ is contained in the zero set of a nonzero homogeneous form of degree at most three if and only if A is projectively equivalent to the Clebsch hexagon. In the Clebsch case the least such degree is two: the quadratic vanishing space is spanned by the defining nonsingular conic equation Q, and the cubic vanishing space is*

$$Q \cdot \mathbb{F}_{11}[X,Y,Z]_1.$$

Computer-assisted proof. For each of the fifteen projective classes in the six-arc census, evaluate the ten cubic monomials on $\mathcal{U}(A)$. The resulting evaluation map has rank ten for every non-Clebsch class and rank seven for the Clebsch class. Thus no nonzero cubic vanishes on a non-Clebsch uncovered locus. A containing form of smaller positive degree could be multiplied by a nonzero monomial to produce such a cubic, so smaller positive degrees are excluded as well; no nonzero constant vanishes on the nonempty locus. For the Clebsch class, the quadratic kernel is generated by Q, and multiplication by linear forms gives the full three-dimensional cubic kernel. □

Corollary 4.4 (Monomial characterization)**.** *Up to monomial equivalence, C is the unique linear $[6,3,4]_{11}$ code of covering radius three whose projective deep-hole syndrome locus is contained in a plane curve of degree at most three. In particular, A_5 is recovered from this coding condition, not assumed as a hypothesis.*

Proof. The columns of a parity-check matrix of such a code form a six-arc A, and covering radius three identifies its projective deep-hole syndrome locus with $\mathcal{U}(A)$. Proposition 4.3 makes A projectively equivalent to the Clebsch hexagon. A projectivity between two unordered parity-check column sets lifts, after choosing column representatives, to a row operation together with a coordinate permutation and nonzero coordinate scalings; these are exactly a monomial code equivalence. The converse is immediate, and Dye's stabilizer theorem gives the final A_5 statement. □

Remark 4.5 (Sharpness of the degree threshold)*.* The cutoff cannot be raised from three to four: the four non-Clebsch classes `C02`, `C11`, `C13`, and `C14` have $d_{\min} = 4$, so the failure is not isolated. For `C02`, the low-degree replay in Table 3 prints a quartic whose rational zero set is exactly the 18-point uncovered locus and verifies the equality point by point. No smoothness, irreducibility, or genus assertion is used here.

Theorem 4.6 (Numerical gap)**.** *The Clebsch class has $|\mathcal{U}(A)| = 12$, while every non-Clebsch six-arc has $|\mathcal{U}(A)| \geq 16$. Equivalently, $|\mathcal{U}(A)| \leq 15$ characterizes the Clebsch class, and equality $|\mathcal{U}(A)| = 12$ is its exact value. If $|\mathcal{U}(A)| \geq 16$, then $\mathcal{U}(A)$ lies on no conic.*

Computer-assisted proof. Any ordered four points of a six-arc form a projective frame, uniquely normalized to $(1,0,0),(0,1,0),(0,0,1),(1,1,1)$. Enumerating the remaining unordered pair subject to the arc condition gives 1548 normalized sets. Each embedded arc has $\binom{6}{4}4! = 360$ ordered-frame choices. For each choice, apply the unique normalizing projectivity, sort the resulting six normalized points, and encode that list lexicographically. Taking the least of the 360 lists removes the frame choice; two arcs have the same least list exactly when a projectivity carries one to the other. This canonical key gives fifteen classes, and the number of distinct normalized presentations of a class with stabilizer G is $360/|G|$. Their extension counts are

$$|\mathcal{U}(A)| \in \{12, 16, 18, 19, 20, 21, 22\},$$

with multiplicities

$$(6, 30, 150, 300, 630, 360, 72)$$

across the 1548 normalized representatives. The six presentations attaining 12 form the single Clebsch class; every other class has at least 16 extension points. Theorem 4.2 supplies the final conic-containment assertion. □

Remark 4.7 (Why this rigidity is not in the extension-locus literature)*.* The set $\mathcal{U}(A)$ is the classical extension locus, studied for sufficiently large arcs through Segre's tangent-envelope method; see Ball and Lavrauw [8, Thms. 39–41]. Those results require arc-size hypotheses far from the six-point regime, while Sadeh's census uses six-arcs for a different classification problem. Neither direction records the conic rigidity detected here.

5 The invariant support bipartition

Label the six coordinates by $0,\ldots,5$. The 20 three-element coordinate supports form two complementary A_5-orbits of size ten. Automorphisms preserve the two classes, but nothing intrinsic chooses one as positive. We use *support bipartition* only as shorthand for this unoriented bipartition. Its ten complementary pairs carry the Petersen graph in Figure 1.

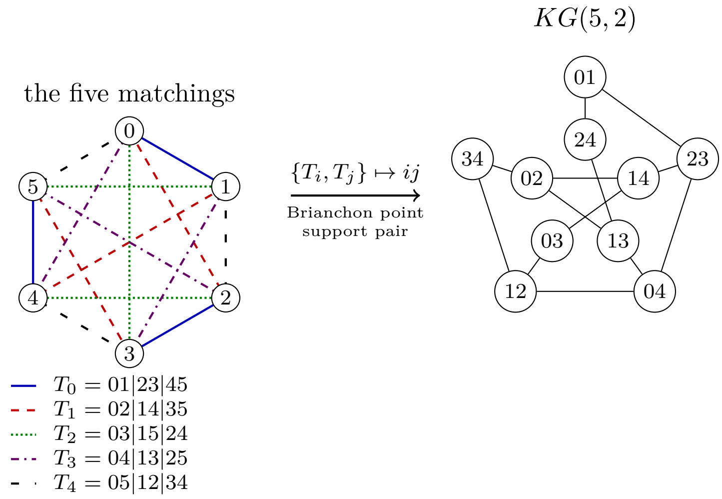

Figure 1: The five-matchings–Petersen dictionary. The line styles encode the five self-polar triangles as perfect matchings of the six coordinates. For each pair $\{T_i, T_j\}$, their union is an alternating six-cycle: its opposite-edge matching gives a Brianchon point, and its alternating vertex classes give a complementary support pair. Both are represented by the Petersen vertex ij; adjacency means that the index pairs are disjoint.

Proposition 5.1 (Brianchon–support dictionary)**.** *Let $\mathcal{T}$ be the five self-polar triangles of Edge and Dye [4, 12], regarded as five perfect matchings, and let $\mathcal{B}(A)$ be the set of ten Brianchon points of the Clebsch hexagon. For distinct $T, T' \in \mathcal{T}$, the union $T \cup T'$ is an alternating six-cycle. Let $M(T,T')$ be the perfect matching joining opposite vertices of this cycle. Let $\beta(T,T') = \{S, S^{\mathsf{c}}\}$ be the unordered pair of alternating vertex classes of the cycle. For example, $T_0 \cup T_1$ is the cycle $0,1,4,5,3,2$, so $M(T_0,T_1) = 05|13|24$ and $\beta(T_0,T_1) = \{\{0,3,4\},\{1,2,5\}\}$.*

The three chords indexed by $M(T,T')$ concur at a point $b(T,T') \in \mathcal{B}(A)$. The maps

$$\binom{\mathcal{T}}{2} \longrightarrow \mathcal{B}(A), \qquad \{T,T'\} \longmapsto b(T,T'),$$

and

$$\binom{\mathcal{T}}{2} \longrightarrow \{\{S,S^{\mathsf{c}}\} : |S| = 3\}, \qquad \{T,T'\} \longmapsto \beta(T,T'),$$

are A_5-equivariant bijections. They identify the ten Brianchon points with the ten complementary support pairs. Under this identification, Petersen adjacency is disjointness of the corresponding two-element subsets of $\mathcal{T}$.

Proof. The opposite-vertex matching is disjoint from T and T' and contains one edge from each of the other three members of $\mathcal{T}$. Hence $\{T, T'\}$ is recovered as the two members of $\mathcal{T}$ disjoint from $M(T, T')$, so the ten triangle pairs give exactly the ten perfect matchings outside $\mathcal{T}$. Applying the alternating-class rule to the five displayed matchings gives ten distinct complementary support pairs, hence all ten such pairs. Edge's Clebsch construction says that the three chords of each such matching concur, at the ten distinct Brianchon points [4, §§19–20,30]. The bipartition bijection and Petersen assertion are the alternating-cycle construction above.

The ten outside matchings give the ten triple concurrences. The remaining fifteen intersections of disjoint chords have multiplicity one. Since every step uses only the five invariant matchings and incidence, both bijections and their compatibility with the support map are A_5-equivariant. □

Corollary 5.2 (The decoder reconstructs the Brianchon configuration)**.** *The ten projective directions having three nearest weight-two errors are exactly the ten Brianchon points. At each such direction the three error supports are pairwise disjoint and hence form a perfect matching of the six coordinates. The ten matchings thereby recovered are precisely the ten perfect matchings outside the five matchings in $\mathcal{T}$.*

Proof. The three weight-two representations of a secant-index-three direction use the three chords through it. Proposition 5.1 identifies the ten triple concurrences with the ten Brianchon points and their chords with exactly the ten matchings outside $\mathcal{T}$. □

Proposition 5.3 (Invariant support bipartition)**.** *Only the bipartition is intrinsic: neither part has a canonical sign. The twenty three-element coordinate supports split into two complementary A_5-orbits $\mathcal{O}_+$ and $\mathcal{O}_-$ of size ten. For every maximum-distance syndrome s, each support determines exactly one weight-three leader of the coset $H^{-1}(s)$; hence that coset has ten leaders supported in each class. Globally, the* 2400 *minimum-weight leaders split as* $1200 + 1200$.

Every monomial automorphism of C preserves this unordered bipartition. The other coset in the exotic normalizer $N_{S_6}(A_5) \cong S_5$ exchanges the two support classes, but its sixty elements are not automorphisms of the displayed code. By Proposition 5.1, the ten complementary support pairs are canonically both the two-subsets of the five self-polar triangles and the ten Brianchon points.

Proof. The degree-six action of A_5 on the twenty three-subsets has two ten-element orbits, exchanged by complementation. It preserves a unique set of five perfect matchings, also preserved by its full S_5 normalizer. The alternating-cycle construction of Figure 1 is therefore equivariant, and the outside normalizer coset exchanges the two support orbits. Fix a maximum-distance syndrome s and a three-support S. The three columns indexed by S are independent because A is an arc, so they give a unique coefficient vector representing s. None of its three coefficients can vanish: otherwise $[s]$ would lie on a secant of A, contrary to maximum distance. The resulting error vector therefore has weight three and is the unique leader with support S. Finally, every monomial automorphism induces a support permutation in A_5, by the exact sequence in Section 3.1, and so preserves each support orbit. The outside normalizer coset swaps the two orbits but cannot preserve C, even after coordinate rescaling, because its support permutations lie outside that quotient A_5. □

6 Why $q = 11$

Call an arc A *conic-filling* if $\mathcal{U}(A) = \mathcal{Q}(\mathbb{F}_q)$ for some nonsingular conic $\mathcal{Q}$. Equivalently, $A \cap \mathcal{Q}(\mathbb{F}_q) = \varnothing$, its chords cover every point outside $A \cup \mathcal{Q}(\mathbb{F}_q)$, and no chord meets $\mathcal{Q}(\mathbb{F}_q)$.

The conic-filling syndrome phenomenon of Sections 3–4 is specific to $q = 11$, even without a symmetry hypothesis on the arc. A universal chord-multiplicity identity and the passant window leave only $q = 9, 11$; the former is excluded by a classical exact-distance problem in the Sylvester graph.

Lemma 6.1 ($q = 9$ polarity graph)**.** *For a nonsingular conic $\mathcal{C} \subset \mathrm{PG}(2, 9)$, let Σ be the graph on the 36 internal points in which*

$$P \sim Q \quad \Longleftrightarrow \quad Q \in P^{\perp}$$

for the conic polarity. Then Σ is the Sylvester graph, with intersection array

$$\{5, 4, 2; 1, 1, 4\}.$$

For distinct internal points P, Q, the line PQ is passant to $\mathcal{C}$ if and only if P and Q are at distance two in Σ, and the graph joining pairs at distance two has clique number

$$\omega(\Sigma_2) = 5.$$

Proof. Counting internal points on polar lines and their intersections gives $b_0 = 5$, $b_1 = 4$, $b_2 = 2$, $c_1 = c_2 = 1$, and $c_3 = 4$, hence the displayed intersection array. This is the Sylvester graph [10, §13.1.2]; uniqueness for this intersection array is proved in [11, Thm. 6].

For distinct internal points P, Q, put $R = P^{\perp} \cap Q^{\perp} = (PQ)^{\perp}$. If PQ is passant, then its pole R is internal and adjacent to both P and Q; since the intersection array has $a_1 = 0$, these vertices are at distance two. Conversely, a common neighbor of P and Q must equal R, so distance two makes the pole of PQ internal and hence makes PQ passant. The final value is $\mathrm{eq}_2(\Sigma) = 5$ in [9, Table 1]. □

Theorem 6.2 (Uniqueness of $q = 11$)**.** *Let q be a prime power. If a six-arc $A \subset \mathrm{PG}(2, q)$ satisfies $\mathcal{U}(A) = \mathcal{C}(\mathbb{F}_q)$ for some nonsingular conic $\mathcal{C}$, then $q = 11$. At $q = 11$ every such A is projectively equivalent to the Clebsch hexagon.*

Proof. Corollary 2.3 gives $q \geq 9$ and $c(A) = (q - 6)(q - 9)$ with $0 \leq c(A) \leq 15$. Thus the only prime-power candidates are $q = 9, 11$: $q = 10$ is not a prime power, and every $q \geq 12$ gives $c(A) \geq 18$.

It remains to exclude $q = 9$. Since every point of $\mathcal{C}$ is uncovered, no chord of A meets $\mathcal{C}$; every chord is therefore passant. A point off $\mathcal{C}$ lies on five passants if it is internal and four if it is external, so the five chords through each vertex force all six vertices of A to be internal. Lemma 6.1 would then make A a six-clique in the distance-two graph of the Sylvester graph, whose clique number is five. This is impossible.

Therefore $q = 11$. The final assertion follows from Theorem 4.2, which puts every matching $q = 11$ representative in the single Clebsch orbit. □

Proposition 6.3 (The Clebsch-family uncovered formula)**.** *Let K be a Clebsch hexagon over $\mathbb{F}_q$. Then*

$$|\mathcal{U}(K)| = q^2 - 14q + 45.$$

If $q \equiv 3 \pmod 4$ and $\mathcal{C}$ is Dye's associated conic, then $\mathcal{C}(\mathbb{F}_q) \subseteq \mathcal{U}(K)$ and

$$|\mathcal{U}(K) \setminus \mathcal{C}(\mathbb{F}_q)| = (q - 4)(q - 11).$$

In particular, among Clebsch hexagons in this congruence class, the associated conic is the complete projective deep-hole syndrome locus if and only if $q = 11$.

Proof. Dye defines a Clebsch hexagon by the occurrence of exactly ten Brianchon points, and his Theorem 1 classifies their existence and projective orbit [12, p. 271; Theorem 1, pp. 275–276]. These points are precisely the off-vertex triple-chord concurrences counted by $c(K)$. Thus $c(K) = 10$, and Theorem 2.2 gives

$$|\mathcal{U}(K)| = q^2 - 14q + 55 - 10 = q^2 - 14q + 45.$$

Dye also shows that the edges of K are non-secants of the associated conic when $q \equiv 3 \pmod 4$ [12, discussion preceding Theorem 6, pp. 281–282]. Hence every rational point of $\mathcal{C}$ is uncovered. Subtracting $|\mathcal{C}(\mathbb{F}_q)| = q + 1$ from the first formula gives

$$q^2 - 15q + 44 = (q - 4)(q - 11).$$

The factor vanishes only at $q = 4$ and $q = 11$, and the stated congruence leaves only $q = 11$. Conversely, Clebsch hexagons exist over $\mathbb{F}_{11}$ by Dye's Theorem 1, and the inclusion above is then an equality because both sets have twelve points. ☐

Remark 6.4 (Three Clebsch-family regimes). The polynomial $q^2 - 14q + 45 = (q - 5)(q - 9)$ vanishes exactly at $q = 5, 9$. The root $q = 5$ lies in the exceptional characteristic, where the associated-conic construction of Definition 2.1 is not available. At $q = 9$, however, Dye's construction applies: $5 = -1$ is a square in $\mathbb{F}_9$, and the characteristic is three. Thus the formula recovers a complete Clebsch six-arc at $q = 9$. Independently, Lemma 6.1 excludes every six-arc over $\mathbb{F}_9$ from conic filling by the Sylvester-graph clique bound. At $q = 11$ the uncovered locus is instead exactly the associated conic.

At $q = 19$, the same icosahedral construction still produces a Clebsch six-arc disjoint from its associated conic, but the covering fails. The twenty conic points remain uncovered [12, p. 281], while Proposition 6.3 gives

$$|\mathcal{U}(A)| = 140 = 20 + 120.$$

Thus $120 = (19 - 4)(19 - 11)$ additional points escape all fifteen secants, and $\mathcal{U}(A)$ lies on no conic. The configuration persists; exact conic filling is what the factorization isolates at $q = 11$.

6.1 The small-arc boundary

The chord moments also put the six-point theorem into a short classification for all smaller neighboring arc sizes.

Theorem 6.5 (Conic-filling loci through eight points)**.** *Let A be a k-arc in* $\mathrm{PG}(2, q)$*, where q is a prime power and $4 \leq k \leq 8$, and suppose that $\mathcal{U}(A)$ is the full set of $\mathbb{F}_q$-rational points of a nonsingular conic. If $k \leq 7$, then*

$$(k, q) = (4, 5) \qquad \text{or} \qquad (k, q) = (6, 11).$$

In the first case A is a projective four-frame; in the second it is a Clebsch hexagon. Conversely, both cases occur. If $k = 8$, then

$$q \in \{13, 17, 19\}.$$

Proof with exhaustive finite verification. For $k = 4$, Corollary 2.3 gives $5 \leq q < 6$, hence $q = 5$. Every four-arc is a projective frame, and for the standard frame a direct substitution gives

$$\mathcal{U}(A) = V(X^2 + Y^2 + Z^2 + XY + XZ + YZ),$$

where $V(f)$ denotes the projective zero set of f; this is a nonsingular six-point conic. For $k = 5$, the theorem gives $|\mathcal{U}(A)| = q^2 - 9q + 21$; equality with $q + 1$ would require $q^2 - 10q + 20 = 0$, which has no integral root. The case $k = 6$ is Theorem 6.2 together with Theorem 4.2.

For $k = 7$, Theorem 2.2 gives

$$|\mathcal{U}(A)| = q^2 - 20q + 120 - n_3.$$

If this equals $q + 1$, then

$$n_3 = q^2 - 21q + 119, \quad n_2 = 105 - 3n_3, \quad n_1 = 3(q^2 - 14q + 42).$$

The conic-filling window and quadratic barrier give $q \geq 11$ and $q^2 - 21q + 84 \leq 0$, hence $q \leq 15$. Thus only the prime powers $q = 11, 13$ remain, with forced spectra $(n_1, n_2, n_3) = (27, 78, 9)$ and $(87, 60, 15)$ respectively. An exhaustive four-frame-normalized enumeration now supplies the finite exclusion. It finds 10232 distinct normalized seven-arcs at $q = 11$, of which 140 have $|\mathcal{U}(A)| = q + 1$, and 53960 at $q = 13$, of which 1680 have that size. Every seven-arc is reached because it contains a four-frame; extending each normalized six-arc by its uncovered points produces every resulting seven-arc exactly three times, according to which non-frame point is deleted. For all 1820 candidates of the required size, the quadratic evaluation matrix has full rank six, so none lies on a conic.

Finally let $k = 8$. The conic-filling window and its quadratic inequality give $q \geq 13$ and $q^2 - 28q + 125 \leq 0$, hence $q \leq 22$. The prime powers in this interval that are odd are exactly $13, 17, 19$. $\square$

Thus conic filling is finite at every fixed arc size; the classification is complete through seven points, and the eight-point problem is reduced to three fields.

7 Verification architecture

Tables 2 and 3 separate the adopted claims, mathematical input, kernel-checked formal proof, and exhaustive finite enumeration. Here "kernel-checked" means that Lean's trusted kernel checks the stated implication; executable replays instead enumerate the finite search space with exact arithmetic. The formal source tree supplied with the release artifact is pinned at commit `bf4fb39ab3c3b06c3f82c2c90d37077d7aa4c520`; it uses Lean 4.32.0-rc1 and Mathlib commit `571b8a8e54219b4d393f75f4b8653fac08197fcc`. Manifest, release, and executable checker paths are relative to this paper root; Lean paths are relative to the formal source root. These tables are the paper's consolidated proof-mode ledger; the theorem statements and proofs do not carry separate "Proof mode" tags. The claim-by-claim manifest is `verification/trust_manifest.json`; the aggregate formal gate is `RelativeConicArcs/Gates/ClebschRigidityTrust.lean`. The clean replay entry point is `verification/verify_release.py`.

From the root of the archival Git-bundle clone, the exact replay commands are

```
cd papers/clebsch-rigidity
nix develop --command python3 verification/verify_release.py \
  --lean-root ../../lean
```

The SHA-256 digest of `verification/checker_outputs.json` is

`d9595a11734de303b21b3a214419e2c76668fa8b97353e44c914ef1077bc0b56`.

The digest of the aggregate Lean gate is `dce51b42bfb384950bbffa756a87c3aabccfda03502d19545988f52aca32cb69`.

Result	Mathematical input	Kernel-checked formalization
A_5 orbits and syndrome conic	Dye; Proposition 3.1	`RelativeConicArcs/Q11A5PointOrbits.lean`; `RelativeConicArcs/Q11Coding.lean`
Decoding and support bipartition	arc–coset dictionary; Edge–Dye triangles	`RelativeConicArcs/Q11DecodingSynthesis.lean`
Rigidity theorem	line bound; chord defect; Dye's Theorems 1 and 3	`RelativeConicArcs/Q11DyeAxioms.lean`; `RelativeConicArcs/Q11DyeConsequences.lean`; `RelativeConicArcs/SixArcDefectBridge.lean`
Universal defect and conic-filling window	two chord moments; passant counts	six-arc specialization only: `RelativeConicArcs/ClebschChordDefect.lean`
Gap and low-degree loci	finite linear algebra over $\mathbb{F}_{11}$	—
Uniqueness of $q = 11$	conic-filling window; cited Sylvester bound	`RelativeConicArcs/ClebschChordDefect.lean`; `RelativeConicArcs/Q9Sylvester.lean`
Clebsch-family formula	Dye's ten Brianchon points and edge criterion	—
Classification through seven; eight-point sieve	universal chord moments and defect bound	`RelativeConicArcs/SmallKChordMoments.lean`; `RelativeConicArcs/SmallKGeometricBridge.lean`

Table 2: Mathematical and formal dependencies of the principal results.

The gap/low-degree result and the Clebsch-family formula have no Lean coverage. Their executable routes in Table 3 are load-bearing checks, not independent supplements to a formal proof.

Result	Exhaustive or independent executable check
A_5 orbits and syndrome conic	`check_rigidity_degenerate_conic.py` (syndrome-conic clause)
Decoding and support bipartition	`check_decoding.py`; `check_chirality.py`; `check_code_automorphisms.py`
Rigidity theorem	`check_rigidity_degenerate_conic.py`; 1548-representative frame-normalized enumeration
Universal defect and conic-filling window	conceptual double counting; no separate executable check
Gap and low-degree loci	`check_global_conic_gap.py`; `check_low_degree_loci.py`
Uniqueness of $q = 11$	`check_small_q_uniqueness.py`; independent $q = 9$ construction
Clebsch-family formula	`check_q19_nonexample.py`
Classification through seven; eight-point sieve	`check_small_k_conic_filling.py`; the eight-point sieve is human algebra

Table 3: Executable checks, separate from the formal dependencies in Table 2.

The formal development axiomatizes exactly two consequences of Dye's Theorem 1: the ten-point Brianchon bound and the classification of equality. The line bound and chord-defect identity are proved independently. At $q = 11$, the finite census also checks the imported numerical conclusion: $|\mathcal{U}(A)| = 12 = 22 - 10$ is equivalent to the maximal value $c(A) = 10$.

8 Conclusion

The universal defect identity makes conic filling finite at every fixed arc size. It explains both the isolation of $q = 11$ and the classification for $4 \le k \le 7$, while the two-sided conic-filling window reduces the eight-point case to three fields. The numerical window is nonempty for every $k \ge 4$ and has width $\binom{k}{2} - (2k-3) = \binom{k-2}{2}$; it is sharp at $k = 4$, where its sole integer is $q = 5$ and conic filling occurs. The Clebsch code realizes another sharp endpoint: covering-radius data recover geometry that was not built into the code, forcing the Clebsch hexagon and hence its A_5 symmetry. A line bound and the defect identity reduce that rigidity to Dye's equality case; enumeration is reserved for the low-degree strengthening, the numerical gap, and the terminal $k = 7$ exclusions.

Closing the three eight-point fields will require further concurrency or extension information. More generally, which odd prime powers in the quadratically widening window admit conic-filling arcs? A second problem is to connect fixed-k rigidity with the large-arc conic-forcing regime. Both ask how far a syndrome locus can reconstruct the projective system that produced it.

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